

TASKS'

1. List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.



1. Faulty Network Cable

Symptom: The computer shows “No Internet” or the network connection keeps disconnecting.

2. Damaged Network Port

Symptom: The device does not detect the Ethernet cable even when the cable is properly connected.

3. Switch Failure

Symptom: Multiple computers connected to the same switch suddenly lose network connectivity.

4. Faulty Network Interface Card (NIC)

Symptom: One computer cannot connect to the network, while other computers work normally.

5. Router Hardware Failure

Symptom: Devices connected to the router cannot access the network or Internet, and the router may show abnormal or no status lights.

2. Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.



```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

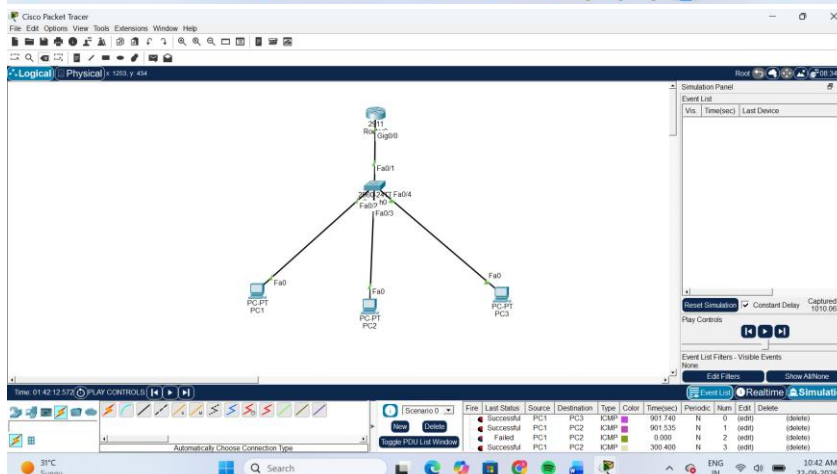
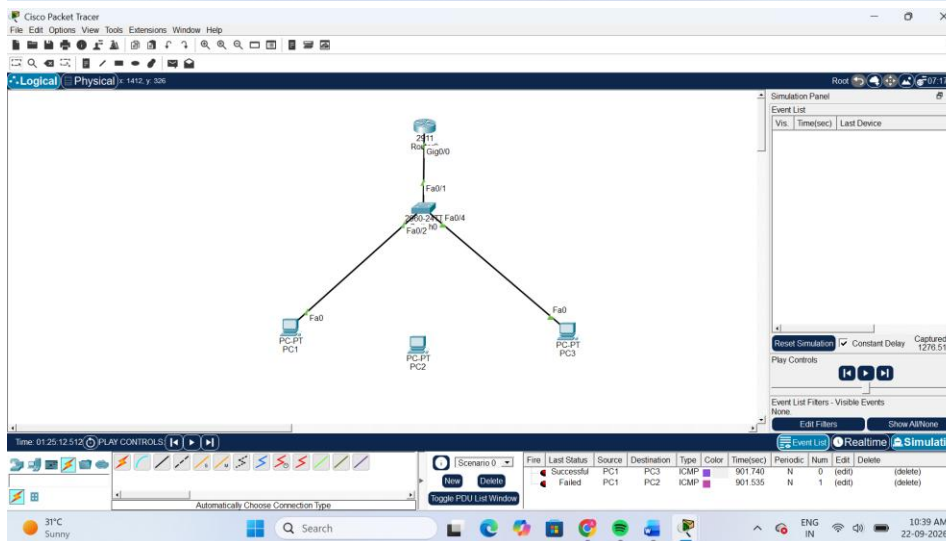
Press RETURN to get started!

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#
  
```



Troubleshooting Steps

1. I created a small network using 3 PCs, 1 switch, and 1 router in Cisco Packet Tracer.
2. I configured the IP addresses of all three PCs and the router.
3. I first tested the network using a Simple PDU (ICMP).
4. The connection between PC1 and PC3 was successful, confirming that the network was working properly.
5. I intentionally disconnected PC2 from the switch to create a network fault.
6. I sent an ICMP packet from PC1 to PC2, which resulted in a Failed status.
7. I checked PC2's IP configuration and found it was correctly configured.
8. I then checked the physical connection and identified that PC2 was disconnected from the switch.
9. I reconnected PC2 to Switch Fa0/3 using a Copper Straight-Through cable.
10. I tested the connection again using Simple PDU.
11. The PC1 → PC2 connection was successfully restored.

Fault Identified

Fault: PC2 was physically disconnected from the switch.

Symptom: ICMP communication from PC1 to PC2 failed.

Solution: Reconnected PC2's FastEthernet0 interface to Switch Fa0/

3. Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.
Hint: Think about checking cables, router status lights, device settings, and running ping or speed tests.



Slow Internet Troubleshooting Checklist

1. Check the cables

Make sure the router's power and Ethernet cables are properly connected and not damaged.

2. Check router status lights

Check the Power, Internet/WAN, and Wi-Fi lights. Make sure they are showing normal status.

3. Restart the router

Turn off the router, wait for about 30 seconds, and turn it on again. Wait for the connection to come back.

4. Check Wi-Fi signal strength

Move closer to the router and check whether the internet becomes faster. A weak signal can cause slow speeds.

5. Check connected devices

Disconnect unnecessary devices because multiple devices using the same Wi-Fi can reduce available bandwidth.

6. Check device Wi-Fi settings

Make sure Wi-Fi is enabled and the device is connected to the correct network. Forget and reconnect to the Wi-Fi if needed.

7. Run a ping test

Open Command Prompt and run ping 8.8.8.8. High response times or packet loss can indicate a network problem.

8. Run a speed test

Use a reliable internet speed test and compare the download and upload speeds with the speed promised by the ISP.

9. Check for background usage

Pause large downloads, cloud synchronization, Windows updates, or streaming applications that may be consuming bandwidth.

10. Test with another device

Connect another phone or laptop to the same Wi-Fi. If all devices are slow, the problem may be with the router or ISP. If only one device is slow, check that device's settings or software.

4. You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9407]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VD\Ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=10ms TTL=118
Reply from 8.8.8.8: bytes=32 time=10ms TTL=118
Reply from 8.8.8.8: bytes=32 time=10ms TTL=118
Reply from 8.8.8.8: bytes=32 time=17ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 10ms, Average = 20ms

C:\Users\VD\Ping youtube.com

Pinging youtube.com [192.178.193.93] with 32 bytes of data:
Reply from 192.178.193.93: bytes=32 time=16ms TTL=197
Reply from 192.178.193.93: bytes=32 time=16ms TTL=197
Reply from 192.178.193.93: bytes=32 time=16ms TTL=197
Reply from 192.178.193.93: bytes=32 time=17ms TTL=197

Ping statistics for 192.178.193.93:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 20ms, Average = 17ms

C:\Users\VD\Ping instagram.com

Pinging instagram.com [57.146.48.34] with 32 bytes of data:
Reply from 57.146.48.34: bytes=32 time=28ms TTL=40
Reply from 57.146.48.34: bytes=32 time=28ms TTL=40
Reply from 57.146.48.34: bytes=32 time=31ms TTL=40
Reply from 57.146.48.34: bytes=32 time=28ms TTL=40

Ping statistics for 57.146.48.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 32ms, Average = 27ms

C:\Users\VD\ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\VD\ipconfig /all

Windows IP Configuration

Host Name . . . . . : AshiDatal
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled . . . . . : No
WINS Proxy Enabled . . . . . : No

Ethernet adapter vEthernet (vm to host):

Connection-specific DNS Suffix . . : 
Description . . . . . : Hyper-V Virtual Ethernet Adapter #2
Physical Address. . . . . : 00-15-5D-10-1C-02
Dhcp Enabled . . . . . : Yes
Autotconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::bc85:8453:4f45:1ff6%21(Preferred)
```